

Hasin Rehana

PhD Candidate, Computer Science · Expected December 2026

Graduate Research Assistant, Department of Biomedical Sciences

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U.S. Permanent Resident

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RESEARCH SUMMARY

PhD candidate in Computer Science (expected December 2026) developing AI methods for biomedical and scientific data, with research spanning large language models, natural language processing, multimodal machine learning, computer vision, and bioinformatics. Dissertation work focuses on transformer/LLM-based biomedical information extraction, ontology-grounded structured prediction, and multimodal clinical modeling. Earlier bioinformatics work includes ontology- and gene-signature-based disease analysis, complemented by current work on biomedical knowledge resources and multimodal data integration. Experienced in Python, PyTorch, Hugging Face, transformer and LLM fine-tuning, model evaluation, and GPU-accelerated computing on Slurm-managed HPC systems.

EDUCATION

Ph.D. in Computer Science

Expected Dec 2026

University of North Dakota, Grand Forks, ND, USA

GPA: 4.00/4.00

- Research focus: Artificial intelligence and machine learning, including large language models, natural language processing, structured information extraction, multimodal learning, computer vision, and scalable model fine-tuning and inference on HPC systems, with primary applications in biomedical and scientific data.

M.Sc. in Computer Science & Engineering

2022

University of Dhaka, Bangladesh

GPA: 3.73/4.00

- Thesis: Plant leaf disease detection using deep learning (object detection on agricultural imagery with Faster R-CNN).

B.Sc. in Computer Science & Engineering

2018

Rajshahi University of Engineering and Technology (RUET), Bangladesh

GPA: 3.63/4.00

- Thesis: Spatial-spectral CNN feature extraction for hyperspectral image classification.

RESEARCH INTERESTS

Artificial Intelligence; Large Language Models (LLMs); Natural Language Processing; Computational Biology and Bioinformatics; Multimodal Learning; Evidence-Grounded and Knowledge-Grounded AI; Computer Vision; AI for Scientific Discovery; Biomedical Informatics.

TECHNICAL SKILLS

- **Artificial Intelligence & Machine Learning:** Large language models, transformer architectures, multimodal learning, convolutional neural networks, supervised learning, model fine-tuning, and model evaluation.
- **Imaging & Vision:** Medical imaging, hyperspectral imaging, object detection, image classification.
- **Natural Language Processing & LLMs:** Named entity recognition, nested NER, relation extraction, structured information extraction, evidence-grounded and ontology-grounded extraction; BERT, Llama, Mistral, Qwen, and GPT-family models.
- **Frameworks & Libraries:** PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV.
- **Infrastructure & Tools:** CUDA, Linux HPC environments, Slurm, Docker, Git, AWS SageMaker, Weights & Biases.
- **Computational Biology & Bioinformatics:** Biomedical knowledge representation, ontology-driven analysis, gene-signature analysis, and multimodal biomedical data integration.
- **Programming:** Python (expert), C++, Java, SQL.

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant — University of North Dakota (Hur Lab)

Jan 2023 – Present

- Develop and evaluate transformer- and large-language-model-based methods for named entity recognition, relation extraction, and structured information extraction across biomedical and scientific corpora.
- Develop multimodal machine-learning approaches that integrate imaging with structured clinical variables to evaluate the contribution of complementary data modalities to predictive modeling.
- Conduct bioinformatics and knowledge-representation research spanning ontology-supported biological knowledge bases, literature-derived molecular mechanisms, and structured biomedical data analysis.

- Developed and evaluated LLM-based pipelines for domain-specific information extraction, including protein–protein interactions, vaccine adjuvants, and associated biomedical mechanisms.
- Developed a multilayer BERT-based architecture for nested named entity recognition, achieving **2nd place in both English and Russian tracks** of the BioASQ12 BioNNE challenge.
- Develop and optimize GPU-accelerated workflows for LLM fine-tuning, inference, and evaluation on Slurm-managed HPC clusters.
- Generate and analyze preliminary LLM results for grant proposals to establish feasibility and support proposed research directions.
- Graduate research appointment supported in part by NIH/NIAID U24 AI171008 (VIOLIN 2.0) and the TRANSCEND Pilot Project supported under NIH/NIGMS P20GM155890.

Clinical Research Study Team Member — Sanford Research

Jun 2025 – Present

- Contribute to an interdisciplinary TAVR study developing computational approaches for vascular-access complication risk prediction using medical imaging and clinical variables.
- Contributed to the development of the successful \$75,000 TRANSCEND Pilot Project proposal supporting this work.
- Participate in data understanding, study design, and computational planning for multimodal integration of imaging and clinical variables.

Participant — NIAID–BRC AI Codeathon, Argonne National Laboratory

Nov 2025

- Project: PubMed Miner ([GitHub](#)) — An AI-Powered Sequence Feature Extraction from Literature (JCVI).
- Contribution: Developed a natural language query translation module that converts free-text user input into structured PubMed search queries for downstream processing, significantly improving tool usability for domain scientists.

Lecturer — Daffodil International University, Bangladesh

Jan 2019 – August 2022

- Taught undergraduate Computer Science courses including Algorithms, Artificial Intelligence, Numerical Methods, and Computer Graphics.
- Mentored undergraduate projects in artificial intelligence, machine learning, deep learning, and computer vision.

GRANTS & RESEARCH FUNDING

- **NIH/NIAID U24 AI171008 — VIOLIN 2.0: Vaccine Information and Ontology Linked kNowledgebase.** *Role: Graduate Research Assistant; doctoral research supported by the award.* Conduct research on LLM-based biomedical information extraction, ontology-supported knowledge-base development, and evidence-grounded extraction from scientific literature.
- **TRANSCEND Pilot Project — AI-Based Vascular-Access Complication Risk Prediction in TAVR.** Supported under NIH/NIGMS award P20GM155890; one-year project award: \$75,000. *Role: Graduate Research Assistant and proposal contributor.* Contributed to Specific Aims, background/significance, study design, preliminary research, and computational planning for multimodal risk prediction using medical imaging and clinical variables.

PUBLICATIONS

* indicates equal contribution

1. **H. Rehana***, N.B. Çam*, M. Basmaci, J. Zheng, C. Jemiyo, Y. He, A. Özgür, J. Hur. (2024). *Evaluating GPT and BERT models for protein–protein interaction identification in biomedical text.* *Bioinformatics Advances*, 4(1), vbae133. **(IF 2.6)**
2. J. Monickaraj*, **H. Rehana***, A. Bernardi, Y. Zheng, T. Lin, L. Liu, A. Madireddi, L. Yeh, J. Zheng, J. Hur, Y. He. (2026). *Vaxjo 2.0: An ontology- and large language model-powered knowledge base of vaccine adjuvants and mechanisms.* *Frontiers in Cellular and Infection Microbiology*, 16, 1763384. **(IF 5.5)**
3. J. Zheng, A.Y. Lin, A. Huffman, . . . , **H. Rehana**, . . . , C. Tao, J. Hur, B. Smith, Y. He. (2026). *VO: The Vaccine Ontology.* *Scientific Data* (Nature Portfolio). doi:10.1038/s41597-026-07298-w. **(IF 7.2)**
4. **H. Rehana**, J. Zheng, F.-Y. Yeh, B. Bansal, N.B. Çam, C. Jemiyo, B. McGregor, A. Özgür, Y. He, J. Hur. (2025). *Leveraging large language models for cancer vaccine adjuvant name extraction from biomedical literature.* *Proceedings of the American Medical Informatics Association (AMIA) Annual Symposium 2025.* (Oral presentation).
5. C. Jemiyo, B.A. McGregor, **H. Rehana**, J. Hur. (2025). *Adverse childhood experiences and chronic health outcomes: evidence from 33 US states in the Behavioral Risk Factor Surveillance System, 2019–2023.* *BMC Public Health*, 25(1), 1650. **(IF 4.4)**
6. **H. Rehana***, B. Bansal*, N.B. Çam, J. Zheng, Y. He, A. Özgür, J. Hur. (2024). *Nested named entity recognition using multilayer BERT-based model.* *CLEF Working Notes.*
7. **H. Rehana**, M.R. Ahmed, R. Chakma, S. Asaduzzaman, M. Raihan. (2021). *A bioinformatics approach for identification of the core ontologies and signature genes of pulmonary disease and associated disease.* *Gene Reports*, 24, 101206.
8. **H. Rehana**, S. Asaduzzaman, R. Chakma. (2021). *3D-2D BCNN based automated feature extraction & classification for hyperspectral imaging.* In *ICIEV & icIVPR.*

9. M.R. Ahmed, **H. Rehana**, S. Asaduzzaman. (2021). *Protein interaction network and drug design of stomach cancer and associated disease: a bioinformatics approach*. *Journal of Proteins and Proteomics*.
10. S. Asaduzzaman, M.R. Ahmed, **H. Rehana**, S. Chakraborty, M.S. Islam. (2021). *Machine learning to reveal an astute risk predictive framework for Gynecologic Cancer and its impact on women psychology: Bangladeshi perspective*. *BMC Bioinformatics*. (IF 4.4)
11. S. Islam, **H. Rehana**, S. Asaduzzaman, S.M. Hossen, R. Hossain, T. Bhuiyan, M.S. Uddin, N. Akter. (2020). *Automated risk prediction by measuring pneumothorax size using deep learning*. In *2020 IEEE Region 10 Symposium (TENSYP)* (pp. 1747–1751). IEEE.
12. **H. Rehana**. (2017). *Bangla handwritten digit classification and recognition using SVM algorithm with HOG features*. *Proceedings of the IEEE Conference on Electrical and Computer Engineering (EICT)*.

PREPRINTS & MANUSCRIPTS IN PREPARATION

- **H. Rehana**, J. Hur. (2026). *Schema-Grounded Multitask Instruction Fine-tuning for Joint Biomedical Named Entity Recognition and Relation Extraction in Pharmacovigilance*. bioRxiv.
- N.B. Çam*, **H. Rehana***, J. Zheng, B. Bansal, Y. He, J. Hur, A. Özgür. (2024). *Ontology-based protein–protein interaction explanation using large language models*. bioRxiv.
- **H. Rehana**, M. Ibrahim, H. Ali. (2023). *Plant disease detection using region-based CNN*. arXiv.
- **H. Rehana**, J. Zheng, Y. He, J. Hur. *CO-PICO/PECO: An Occupational Condition-based PICO/PECO Framework for Structuring Occupational Safety and Health Evidence using Generative Pretrained Transformers*. (Manuscript in preparation).
- **H. Rehana**, Y. He, J. Hur. *Evidence-Grounded Evaluation of Fine-Tuned Large Language Models for Vaccine Adjuvant Extraction from Infectious Disease Literature*. (Manuscript in preparation).

ORAL PRESENTATIONS

- **H. Rehana**, Y. He, J. Hur. *Schema-Grounded Multitask Instruction Fine-tuning for Joint Biomedical Named Entity Recognition and Relation Extraction*. International Workshop on Vaccine and Drug Ontology Studies (VDOS), International Conference on Biological and Biomedical Ontology (ICBO), 2026.
- **H. Rehana**, J. Zheng, Y. He, J. Hur. *Condition-based occupational PICO/PECO and its applications for LLM extraction and ontological modeling of structured occupational safety and health*. International Conference on Intelligent Biology and Medicine (ICIBM), 2026. (Flash talk; Accepted with Travel Award)
- **H. Rehana**, J. Zheng, F.Y. Yeh, B. Bansal, N.B. Çam, C. Jemiyo, B. McGregor, A. Özgür, Y. He, J. Hur. *Leveraging large language models for cancer vaccine adjuvant name extraction from biomedical literature*. American Medical Informatics Association (AMIA) Annual Symposium 2025. (**Travel Award**)
- **H. Rehana**, J. Zheng, J. Hur and Y. He. *PICO/PECO-Guided Large Language Model Extraction and Ontological Modeling of Structured Occupational Safety and Health Evidence from Biomedical Literature*. 16th International Conference on Biological and Biomedical Ontology, Nov 9, 2025. (Virtual)
- **H. Rehana**, J. Zheng, L. Yeh, B. Oğuzoğlu, A. Özgür, Y. He, and J. Hur. *Automated Extraction and Classification of Vaccine Adjuvant Mechanisms Using Large Language Models*. 14th International Workshop on Vaccine and Drug Ontology Studies (VDOS), Nov 7, 2025. (Virtual)
- **H. Rehana**, J. Zheng, L. Yeh, B. Bansal, N.B. Çam, C. Jemiyo, B. McGregor, A. Özgür, Y. He, J. Hur. *Identifying cancer vaccine adjuvants in biomedical literature using large language models*. Great Lakes Bioinformatics Conference (GLBIO), May 12–15, 2025.
- **H. Rehana**, Y. He, J. Hur. *Cancer vaccine adjuvant name recognition from clinical trial data using large language models*. 13th International Workshop on Vaccine and Drug Ontology Studies (VDOS), July 22, 2024. University of Twente, Enschede, The Netherlands. (Virtual)
- **H. Rehana**, J. Zheng, Y. He, J. Hur. *ChatGPT to identify drug-drug interactions from biomedical texts*. 14th ICBO – VDOS Workshop, August 28–September 1, 2023. Brasília, Brazil. (Virtual)

SELECTED POSTERS

- **H. Rehana**, J. Hur. *Schema-Grounded Multitask Instruction Fine-tuning for Joint Biomedical Named Entity Recognition and Relation Extraction in Pharmacovigilance*. Women in Machine Learning (WiML) Workshop @ NeurIPS 2026. (**Accepted Poster**)
- **H. Rehana**, N.B. Çam, M. Basmaci, A. Özgür, Y. He, J. Hur. *Identifying protein–protein interaction from biomedical text using generative pre-trained transformer*. University of North Dakota Cybersecurity Awareness and Research Symposium (CARS), October 30–31, 2023. Grand Forks, ND. (**1st place; Best Poster**)
- **H. Rehana**, H. Row, J. Hur, M. Kim, C. Dyke. *A Literature-Grounded Artificial Intelligence Framework for Vascular Access Complication Risk Modeling in Transcatheter Aortic Valve Replacement*. TRANSCEND Symposium 2026. (University of North Dakota, Grand Forks; Sanford Health, Fargo.)

- **H. Rehana**, N.B. Çam, M. Basmaci, A. Özgür, Y. He, J. Hur. *Leveraging large language models for extracting protein–protein interactions from biomedical corpora*. South Dakota State University Data Science Symposium, February 5–6, 2024. Sioux Falls, SD. (**Travel Award**)

AWARDS & HONORS

- **Travel Stipend**: NIAID BRC AI Codeathon 2025 (Argonne National Lab)
- **Conference Travel Awards**: AMIA Annual Symposium 2025 (University of North Dakota) ; Data Science Symposium 2024 (South Dakota State University); International Conference on Intelligent Biology and Medicine 2026 (ICIBM)
- **Challenge Award**: BioASQ12 BioNNE Task, 2024 — 2nd place in both English and Russian Nested Named Entity Recognition tracks.
- **Best Poster Award**: Cyber Awareness and Research Symposium 2023 (University of North Dakota)
- **ICT Master’s Research Fellowship**: ICT Division, Government of Bangladesh (2021–2022)

CERTIFICATIONS & COMPLIANCE

- **Human Subjects & Clinical Research Compliance (CITI Program)**: Active certifications (2025–2028) in Biomedical Research Investigators, HIPAA for Researchers, Good Clinical Practice (GCP – U.S. FDA Focus), Responsible Conduct of Research, and Data or Specimens Only Research.
- **Clinical Data Privacy & Governance**: Trained in HIPAA-compliant handling of sensitive patient data, clinical records, and biomedical research datasets.
- **Professional ML Training (NVIDIA Deep Learning Institute)**: Fundamentals of Deep Learning; Building Transformer-Based Natural Language Processing Applications; Applications of AI for Anomaly Detection.

SERVICE & MEMBERSHIPS

- **Reviewer**: Scientific Reports (Springer Nature) — Reviewed 2 manuscripts.
- **Memberships**: American Medical Informatics Association (AMIA), Institute of Electrical and Electronics Engineers (IEEE), Association for Computing Machinery (ACM), International Society for Computational Biology (ISCB).